Participant's Section

I. Use Case Short Title

Treasury Pulse : Short-term Liquidity Manager

II. Problem Statement & Background

In a global organization, "Cash" is not just one number in one bank account. It is thousands of different balances held in various currencies (USD, EUR, JPY, etc.), across different banks, in different time zones, and owned by different legal entities (subsidiaries).

Currently, the process is a fragmented, manual "data collection marathon." Every morning (or end-of-day), treasury staff must:

1. Log in to multiple banking portals to download statements for each entity.
2. Download internal ERP (Enterprise Resource Planning) reports to see scheduled upcoming payments (invoices, payroll, taxes).
3. Consolidate this data into massive Excel spreadsheets. This involves manually "cleaning" the data, converting different currencies into a single "reporting currency" using manual exchange rates, and ensuring that a payment sent from the US entity is correctly accounted for when it arrives in the UK entity.
4. Reconcile discrepancies: Manually checking why the bank says there is $1M but the internal system says there is $1.2M.

Treasury does this to answer three critical questions every single day:

- Solvency: "Do we have enough cash in this specific currency in this specific country to pay our employees and suppliers tomorrow?" (If not, we hit expensive overdrafts).
- Optimization: "We have $50M sitting idle in a EUR account doing nothing. Can we move it to a higher-interest USD account or use it to pay off a debt?" (If we don't track it, we lose interest income).
- Risk Mitigation: "If the Euro drops by 5% against the Dollar tomorrow, how much money will our European entities lose?"

Because this is done in Excel:

- It is "Rear-view Mirror" reporting: By the time the spreadsheet is finished, the data is already 12–24 hours old. You are looking at where the money was, not where it is going.
- Human Error is inevitable: One broken Excel formula or one missed bank statement can lead to a multi-million-dollar mistake.

- Complexity is unmanageable: As the number of entities and currencies grows, the spreadsheet becomes too heavy and slow to use for real-time decision-making.

Liquidity Gap Calculation

The Gap (G) is the difference between the Projected Cash Position (P) and the Minimum Threshold (T).

Formula:

$G = P - T$

Where P (Projected Cash Position) is calculated as:

$P = B_{prev} + (Total_{Inflows} - Total_{Outflows})$

Variable Definitions:

- G (Gap): The resulting liquidity status.
- P (Projected Position): The predicted cash balance at a specific time.
- T (Threshold): The minimum safety buffer required for that entity/currency.
- B_prev (Previous Balance): The closing balance from the previous period.
- Total_Inflows: The sum of all expected incoming cash for the period.
- Total_Outflows: The sum of all expected outgoing cash for the period.

Gap Interpretation:

- If $G > 0$: The entity has a Surplus (Cash is available).
- If $G < 0$: The entity has a Shortfall (Liquidity risk/potential overdraft).

III. Objective

The primary objective is to architect and develop a centralized Liquidity Command Center that transitions treasury operations from a manual, "rear-view mirror" approach to an AI-assisted, proactive management model.

The project aims to achieve three strategic pillars:

1. Operational Transformation (From Manual to Automated)

- Automated Data Aggregation: Eliminate the "data collection marathon" by automating the ingestion and cleaning of fragmented transaction data from multiple entities and currencies.
- Single Source of Truth: Provide a unified, real-time view of global cash positions, eliminating discrepancies between banking portals and internal ERP systems.

2. Predictive Risk Management (From Reactive to Proactive)

- Anticipatory Liquidity Visibility: Move beyond historical reporting to predictive modeling, providing a 30-day rolling window of expected cash positions.
- Automated Early Warning Systems: Implement a dynamic Alert Matrix that identifies potential solvency risks (shortfalls) and optimization opportunities (surpluses) before they occur.

3. Strategic Decision Intelligence (From Data to Action)

- Quantified Scenario Modeling: Enable treasury officers to simulate "What-If" scenarios—such as FX volatility, payment delays, or capital expenditure shifts—to understand their impact on liquidity before committing to actions.

- AI-Powered Advisory: Leverage Generative AI (RAG) to provide natural-language explanations for forecast anomalies and suggest optimized treasury workflows (e.g., cash pooling, delaying payments, or investing idle funds).

IV. Definition of Done

Phase	Deliverables
Core	CSV upload, transaction parsing, Dashboard with daily cash position, 30-day forecast, entity filter, liquidity gap table, alert matrix, what-if slider, User Access Control, Reporting Engine.
Advanced	AI chatbot (forecast rationale, Cash surplus, action plans, Charts), what-if engine, recommendation engine

V. Expected Outcomes

Outcome	Business Value
Daily cash position per entity/currency	Eliminate manual tracking; real-time visibility
30-day cash balance forecast with confidence intervals	Proactive cash management; reduce overdraft costs
Liquidity gap tracking vs configurable thresholds	Prevent liquidity shortfalls before they occur
Reporting	Scheduled Reports to be auto triggered at start & end of Day.
AI-powered treasury action recommendations	Data-driven decisions; free staff for higher-value work
What-if scenario analysis	Quantify impact of payment delays, capex changes, FX shocks

VI. Solution Requirements

Requirement	Specification
Forecasting	Time-series model (Prophet/SARIMA) - 30-day forecast per entity/currency with confidence intervals
Alert Matrix	Configurable thresholds per entity/region/currency. Per-currency limits: EUR 50k, USD 60k, JPY 5M, GBP 40k. Three levels: Warning (5-10%), Critical (<5%), Emergency (negative). Absolute or percentage based.
What-If Analysis	Simulate payment delays (+1 to +5 days), capex postponement, FX shocks, Interest Rate Shock[Loan/Deposits], CounterParty Rating. Compare baseline vs scenario
Reporting	Scheduled Reports to be auto triggered at start & end of Day.
AI Chatbot	Forecast rationale, surplus identification, treasury action recommendations. RAG for data-grounded responses.

VII. Detailed breakdown of Tasks

Basic Tasks - Must Have

i. Data Ingestion

- Build REST API with file upload and transaction parsing. Calculate daily cash position per entity/currency. Expose data endpoints for frontend consumption.

ii. User Interface

- Set up web application with navigation and layout. Display classified transactions in a filterable table. Build entity-level dashboard with basic charts. Display daily cash position and 30-day forecast per entity/currency. Show liquidity gap vs configurable thresholds.

iii. Forecasting

- Implement time-series forecasting model for 30-day cash balance prediction. Generate forecasts with confidence intervals.

iv. Alert System

- Implement configurable alert matrix with three severity levels. Display alerts on dashboard with breach indicators.

v. What-If Analysis

- Build scenario slider to simulate payment timing variations. Simulate capex postponement impact on forecast.

vi. Reporting

- Build basic scheduler for SOD/EOD reports. Support PDF export.

vii. User Access Control

- Implement role-based access (Viewer, Admin). Enforce entity-level data isolation.

Intermediate Tasks - Must Have

i. Forecasting

- Model seasonality patterns: daily, weekly, monthly, quarterly cycles. Incorporate Counterparty Risk Ratings into forecast accuracy weighting (A=high confidence, D=low confidence in payment timeliness). Calibrate forecast model bias using historical cash-position snapshots

ii. Dashboard

- Stacked Area Chart: Hourly inflows vs outflows visualization with entity/currency drill-down. Liquidity Gap Chart: Projected position with threshold line, breach indicators, and real-time gap calculation (Gap = Projected Position - Threshold). Currency Breakdown Donut Chart: Position distribution across all currencies with percentage labels. Position Trend Table: Time-series view of cash positions with sparkline trend indicators. Alert Bands: Visual zones on charts – green (position 10%+ above threshold), amber (5-10% above), red (<5% above or negative). Entity selector with hierarchical drill-down (Entity → Region)

iii. Alert System

- Build alert matrix UI with per-entity/region/currency threshold configuration. Implement per-currency limits: EUR 50k, USD 60k, JPY 5M, GBP 40k. Implement alert evaluation engine with hourly gap checks. Three configurable alert levels warning: Position within 5-10% of threshold. Critical: Position within 0-5% of threshold. Emergency: Position below threshold or negative. Create alert banner on breach + alert history log with timestamps, severity, and resolution status

iv. What-If Scenario Analysis

- Payment Timing Slider: Adjust payment delays (+1 to +5 days) and see impact on forecast and liquidity gap. Capex Postponement Slider: Adjust capital expenditure timing and see impact on cash position. FX Rate Slider: Adjust exchange rates (±5%, ±10%, ±15%) to simulate FX shocks and see impact on multi-currency positions and FX gain/loss. Interest Rate Slider: Adjust rates (±50bps, ±100bps, ±200bps) applicable to loan and deposit transactions, impacting interest income/expense in the forecast. Compare baseline vs scenario results side-by-side in dashboard. Quantify impact on: liquidity gap, overdraft costs,

surplus deployment opportunities. Counterparty Rating Shift: Simulate counterparty rating downgrades and see impact on payment timeliness assumptions

v. Reporting

- Build configurable reporting scheduler with user-facing UI. SOD Briefing (Start-of-Day): Expected opening position, upcoming payments (next 24-48h), forecast outlook, overnight alerts. EOD Report (End-of-Day): Position summary, forecast, alerts triggered, liquidity gap status for business day. Add CSV export format alongside PDF. Admin-configurable recipients (email distribution list) and delivery schedules. Users enable/disable toggles per report type (SOD/EOD)

vi. User Access Control

- Expand roles: Analyst, Treasury Manager, Admin. Analyst: Entity-level data + what-if scenario analysis + report generation. Treasury Manager: Region-level aggregated view (APAC, EMEA, NAMER) + entity drill-down. Admin: Full access + report scheduler configuration + alert matrix management + user role assignment. Implement session-based role validation on API endpoints

Advanced Tasks - Good to Have

i. Forecasting

- Forecast Rationale: Explain forecast drivers to users (e.g., "Shortfall driven by 3 large loan repayable on Day 3"). Surplus Identification: Auto-flag entities with position above threshold + 20% as deployable surplus. Recommendation Engine: Score treasury actions by expected liquidity impact: Accelerate collections. Delay non-urgent payments. Invest surplus in short-term instruments. Transfer cash between entities/currencies. Utilize cash pooling.

ii. Chatbot Interface

- Build conversational AI (LangChain/OpenAI/Claude) integrating forecast, gap, what-if, surplus, and alert data. Implement RAG (Retrieval-Augmented Generation) for data-grounded responses. Handle natural language queries: e.x. "Why is there a shortfall in EUR next week?". "What-if we delay payments by 3 days?". "What's my surplus position in APAC?" "What action plan do you recommend?". Enable follow-up questions and drill-down to entity/currency level. Generate structured action plans with prioritized recommendations. Chatbot should be able to respond with charts if requested.

iii. Performance Benchmarking

- Forecast Accuracy: Measure MAPE and RMSE against historical actuals. Response Time: Dashboard load time, forecast generation time, chatbot response time. Data Processing Speed: Parsing and processing time for 10K, 50K, and 100K transactions. Establish baseline metrics and track improvements across iterations.

Architecture & Documentation: Deliver a system architecture diagram, data schema diagram, and a demo video ≤ 2 minutes showing end-to-end functionality.

VIII. Input

Source	Format / Details

Transaction Data	CSV/Excel - 12 months, 5000+ records, columns (transaction_id, date, entity, region, currency, amount, direction, product_type, etc.)
Cash-Position Snapshots	Compute Daily end-of-day balances per Entity.
Business Rules	Configured via admin UI - liquidity thresholds, overdraft limits, funding thresholds

IX. Guardrails

Out of Scope / Do Not Do
Production-grade security (OAuth, vault, etc.)
External cloud services requiring internet access
Using production or personally identifiable data - synthetic/anonymized data only
Perfection over functionality - aim for working demo, not production code

X. Dataset

Synthetic dataset (50000 transactions, 12 months, 8 entities, 10+ currencies)

- Refer CSV

Data Dictionary

Column	Description
transaction_id	Unique transaction identifier
transaction_date	Date on which transaction occurred
Entity	Legal entity/subsidiary
Region	Geographic region
Country	Country of entity
City	City of entity
account_type	Type of bank account
account_number	Bank account identifier
Currency	Entity base currency (reporting)
original_amount	Amount in base/source currency

Direction	Transaction direction flow from Bank/Entity perspective [Inflow/Outflow]
product_type	Transaction product category
Category	Cash flow classification
Frequency	Payment frequency
is_recurring	Whether transaction repeats on a timely basis or is one time
payment_method	Settlement channel
Counterparty	Sender/Receiver name
settlement_status	Current settlement state
due_date	Expected settlement date
entity_code	4-char entity abbreviation
Department	Cost center classification
counterparty_risk_rating	Defines confidence in Transaction timeliness & success.
origin_currency	currency of transaction sender
destination_currency	currency of transaction recipient
is_cross_border	Whether the transaction is Cross-borde/country
Fx_rate	Exchange rate from origin_currency to destination_currency at transaction date
maturity_date	Maturity date for financial instruments that have a maturity [e.x. Loan, Deposit]

XI. Reference - Glossary of Business and Functional Terms

Treasury	The department within a bank or corporation is responsible for managing liquidity, cash flow, funding, financial risk (FX, interest rate), and banking relationships. Treasury ensures the organization has sufficient cash to meet obligations while optimizing surplus funds for maximum return.
Liquidity Gap	Difference between projected cash position and minimum required threshold.
Cash Surplus	Entity positions exceeding threshold by more than 20%, indicating deployable idle cash.
EOD Report	End-of-Day summary report showing position, forecast, alerts for the business day.
SOD Briefing	Start-of-Day briefing report showing expected position, upcoming payments, forecast outlook.
RAG	Retrieval-Augmented Generation - AI technique grounding LLM responses in actual system data.

What-If Analysis	Scenario simulation that adjusts input parameters to measure impact on forecast.
Alert Matrix	Configurable threshold system with Warning, Critical, Emergency levels.
Entity	A legal entity or subsidiary (e.g., BNPPARIBAS_PARIS, NY, MUMBAI)
Forecast Horizon	The time period covered by the forecast (30-day)
FX Risk	Exposure to foreign exchange rate fluctuations affecting multi-currency cash positions.
MAPE	Mean Absolute Percentage Error - measures forecast accuracy as a percentage.
RMSE	Root Mean Square Error - measures average magnitude of forecast errors.

XII. Technical Stack Reference

Layer	Technology
Frontend	React / Angular
Backend	FastAPI / Flask (Python) or Spring Boot (Java)
Forecasting	Prophet, SARIMA, scikit-learn
AI/ML	LangChain, XGBoost, OpenAI API
Visualization	Plotly, D3.js, Chart.js
Database	SQLite / PostgreSQL
Chatbot	LangChain + OpenAI / Claude
Streaming (optional)	WebSockets, Server-Sent Events
Vector DB (optional)	Chroma, FAISS, Weaviate